PENETRATION TESTING REPORT

Footprinting & Network Scanning Phases

W2-PM1 + W2-PM5 | CYBERSECURITY & ETHICAL HACKING | NETWORKWALKS

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College	—
Program / Batch	B084 - Networkwalks
Date	13 November 2026
Modules completed	W2-PM1 (Footprinting with Multiple Kali Tools) W2-PM5 (Network Scanning with Zenmap)
Target	1. networkwalks.com (the lab target assigned in the course) 2. My own local LAN network
Permission secured?	Yes. The website is the assigned course lab target, and the LAN is my own network.
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery

1. Liability Disclaimer

I performed these activities only on the lab target assigned in my course and on a network that I own. This material is for education and research only. Do not use anything here to break the law. Unauthorised access is a crime in most countries, even when nothing is damaged. Every action you take is your own responsibility.

2. Introduction

This report covers two Week 2 practicals. In W2-PM1, I footprinted the domain networkwalks.com using six Kali Linux tools. In W2-PM5, I used Zenmap on Windows to discover live hosts on my own local network.

Together they show how a tester first gathers public information about a target, then maps which hosts are alive on a network. For every step I record the command used, the result, a screenshot as evidence, and a short note on why it matters from an attacker's point of view.

3. Tools Used

Tool	Purpose
Kali Linux & Windows	Operating systems used for the activities
whois	Find domain registration details (registrar, dates, name servers)
whatweb	Fingerprint web technologies (server, CMS, plugins, IP)
nslookup	Resolve the domain name to its IP address
curl -I	Read the HTTP response headers
wafw00f	Detect whether a Web Application Firewall protects the site
dnsrecon	Enumerate DNS records (NS, MX, SPF, TXT, SRV)
Zenmap (Nmap GUI)	Scan the local subnet for live hosts, IPs and MAC addresses

Tool	Purpose
Windows CMD (ipconfig)	Find my local IP address, subnet mask and MAC address

4. Activities Performed

4.1 Footprinting & Reconnaissance (W2-PM1)

All six tools only read public information. None of them attacks the target.

Task 1: whois

```
$ whois networkwalks.com
```

The record shows the domain registered through GoDaddy. It was created on 2019-11-06 and is set to expire on 2027-11-06. The name servers are NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM, so the hosting provider is HostGator.

Why it matters: registrar details, dates and name servers reveal the hosting provider and help with social engineering and planning.

Task 2: whatweb

```
$ whatweb networkwalks.com
```

The site redirects from HTTP to HTTPS and runs on Apache. It uses WordPress 7.0.4 with WP Download Manager 3.3.58, plus Bootstrap and jQuery 3.7.1. The output also exposed the server IP 192.232.216.135 and the email address info@networkwalks.com.

Why it matters: exact software versions can be compared with vulnerability databases to look for known issues.

Task 3: nslookup

```
$ nslookup networkwalks.com
```

The domain resolves to 192.232.216.135.

Why it matters: knowing the real IP lets an attacker scan the server directly and map related infrastructure.

Task 4: curl -I

```
$ curl -I https://networkwalks.com
```

The response was HTTP/2 200. The headers show server: Apache, a set-cookie value, an x-nginx-cache: WordPress header and a link header that exposes the WordPress REST API at /wp-json/.

Why it matters: headers reveal the web server, caching setup and hidden endpoints without loading the full page.

Task 5: wafw00f

```
$ wafw00f networkwalks.com
```

The site is behind a ModSecurity (SpiderLabs) Web Application Firewall.

Why it matters: knowing a WAF is present tells an attacker that simple attempts will be blocked or logged.

Task 6: dnsrecon

```
$ dnsrecon -d networkwalks.com
```

DNSRecon found 8 records. They include the SOA and name servers on HostGator, DNS software version Bind 9.16.23, an MX record for mail.networkwalks.com (192.232.216.135), an SPF TXT

record, a Google site-verification TXT record, and several cPanel SRV records (_autodiscover._tcp).

Why it matters: DNS records map the mail and hosting setup and expose software versions.

Note: values above come from the W2-PM1 lab output. Replace them with the exact results from my own Kali terminal if they differ.

4.2 Network Scanning with Zenmap (W2-PM5)

Task 1: Download and install Zenmap

I downloaded the latest stable Windows installer from the official site (https://nmap.org/download.html) and installed it with the default options, keeping Zenmap and Npcap selected. The Nmap - Zenmap GUI icon then appeared on my desktop.

Task 2: Local IP address and LAN subnet

I ran ipconfig in Command Prompt.

Item	Value
IPv4 address	[your IPv4 address]
Subnet mask	[your subnet mask]
Subnet used for scan	[your subnet, e.g. 192.168.1.0/24]
Default gateway	[your gateway]

I also ran ipconfig /all to get the Physical (MAC) address of my active adapter.

Task 3: Live hosts

I opened Zenmap as administrator, entered my subnet as the target, chose the Ping scan profile and clicked Scan.

```
nmap -sn [your subnet]
```

Result:

```
Nmap done: 256 IP addresses (1 host up) scanned in 11.81
seconds
```

Task 4: Number of live hosts

1 host is live ([say which one, e.g. my own PC]). Only one host responded, which can happen because of network configuration such as client isolation, or because other devices do not answer pings.

Task 5: IP addresses of live hosts

Task 6: MAC addresses of live hosts

Task 7: Topology

In Zenmap I opened the Topology tab, turned on the legend and used Save Graphic to save the topology as Zenmap_Topology.pdf on my desktop. With one live host, the topology shows only my PC (localhost).

5. Risk Analysis / Impact

#	Finding	Evidence	Potential impact	Risk
1	Web technology and version information exposed	WhatWeb identified WordPress 7.0.4 and WP Download Manager 3.3.58	Attackers may use exposed versions to look for known issues	Medium
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Shows the network location of the web service	Low
3	HTTP technical information exposed	Curl headers showed Apache, caching info and /wp-json/	Helps fingerprinting and further enumeration	Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals part of the security architecture	Low
5	DNS infrastructure and software version exposed	DNSRecon showed MX, SPF, SRV records and Bind 9.16.23	Helps build a wider profile of mail and DNS setup	Medium
6	Contact email exposed on the website	WhatWeb listed info@networkwalks.com	Can be used for phishing or social engineering	Low
7	Local host discoverable on the LAN	Zenmap ping scan found 1 live host	Devices on the same network can see that the host is online	Low

Risk level key: Critical, Medium, Low.

These are observations, not confirmed vulnerabilities. Only information gathering and host discovery were done. No exploitation or vulnerability validation was performed, and a visible software version does not by itself mean a system is vulnerable. Further authorised testing would be needed to confirm any real issue.

6. Recommendations

1. **Review publicly exposed technology information.** Check what the website reveals about its CMS, plugins and versions.
2. **Keep software updated.** Update WordPress, plugins and server software and check them against security advisories.
3. **Review HTTP headers.** Remove unnecessary technical details from server responses.
4. **Review DNS records regularly.** Keep only the records and services that are needed, and keep DNS software updated.
5. **Keep the WAF enabled and tuned.** ModSecurity already blocks simple attacks, so keep its rules current.
6. **Limit exposed contact details.** Avoid publishing plain email addresses where possible.
7. **Scan your own network regularly.** Check which devices are connected and investigate any unknown device.
8. **Maintain network documentation.** Keep topology and device lists up to date.
9. **Test only with authorisation.** Reconnaissance and scanning must stay within an approved scope.

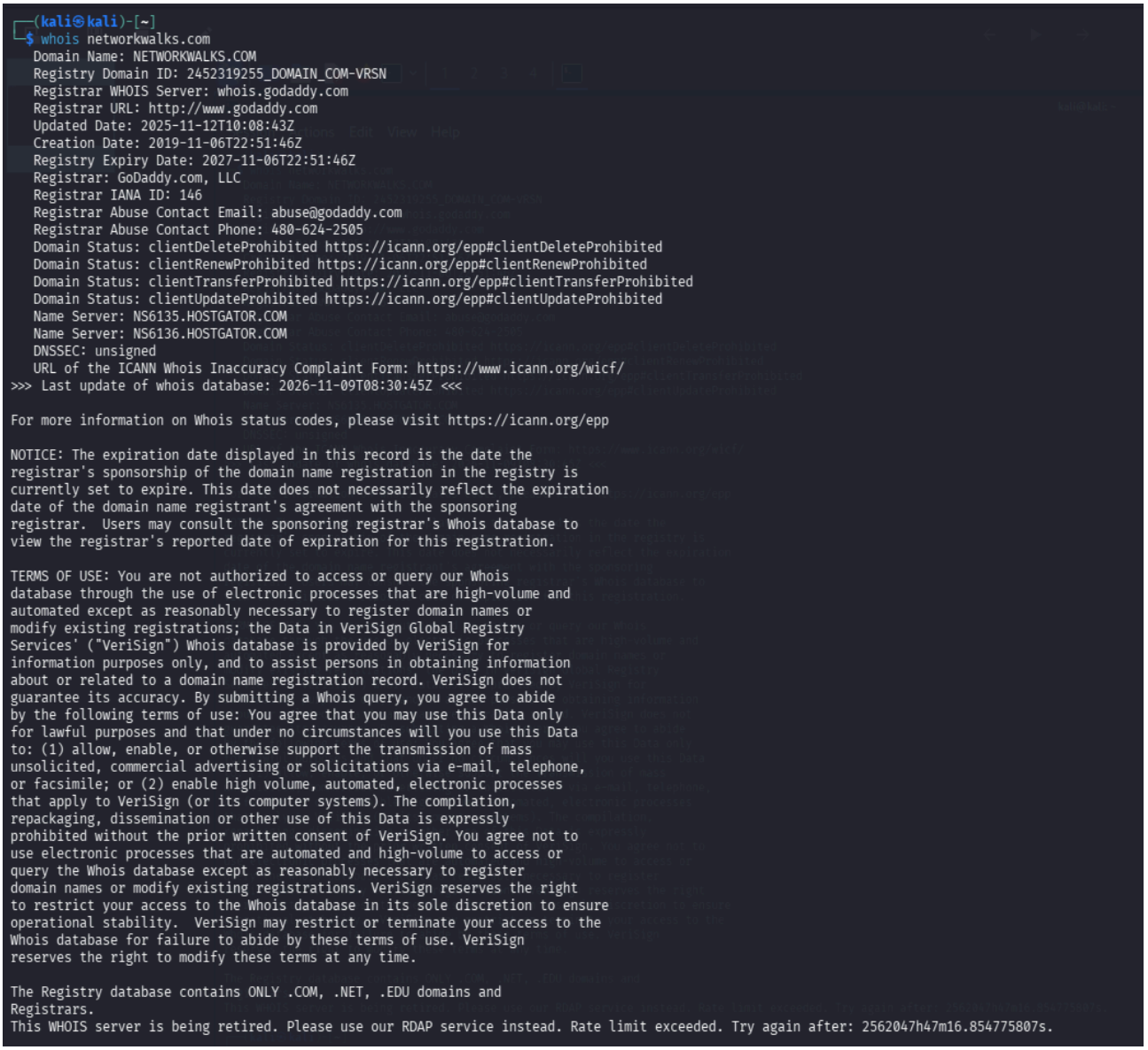
7. Conclusion

In Week 2 of my Cybersecurity & Ethical Hacking program, I completed practical work in footprinting and network scanning. With six Kali tools I learned how whois, whatweb, nslookup, curl, wafw00f and dnsrecon each reveal a different part of a target, from ownership and hosting to software versions, firewall use and DNS setup. With Zenmap I learned how to find my IP address and subnet, run a ping scan, read live hosts and save a network topology.

My own scan found 1 live host, which differs from the 4 hosts in the guide's example, because results depend on how the network is configured. The exercises showed that a lot can be learned before any attack is attempted, and that a good report explains what was done, what was found, what it means, what risk it creates and how to reduce it. I also learned that all of this must stay within an authorised scope.

8. Evidences Collected

Screenshots for each task are placed under the matching task heading above. The topology PDF (Zenmap_Topology.pdf) is submitted as a separate file with this report.



```
┌──(kali㉿kali)-[~]
└─$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&amp;pidnVar2=505116&amp;prtVar2=7&amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.4], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&amp;pidnVar2=505116&amp;prtVar2=7&amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
```

```
┌──(kali㉿kali)-[~]
└─$ nslookup networkwalks.com
Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
```

```
┌──(kali㉿kali)-[~]
└─$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=0466110da3e309b5bd1da216a14d7894; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Mon, 09 Nov 2026 08:58:36 GMT
server: Apache
```

```
┌──(kali㉿kali)-[~]
└─$ wafw00f networkwalks.com

                ( Woof! )
                 ,,
            .-.  _
           ()  ; ⊨═══════)
          / (
     (   /  )
      \(_)_))

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                     ) (_
                    (  |__|
                   .)|__|
                  (  |__|
                   . |__|
                     |__|

                ~ WAFW00F : v2.4.2 ~
    The Web Application Firewall Fingerprinting Toolkit

[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2
```

```
┌──(kali㉿kali)-[~]
└─$ dnsrecon -d networkwalks.com
2026-11-09T04:08:06.942520-0500 INFO Starting enumeration for domain: networkwalks.com
2026-11-09T04:08:06.943313-0500 INFO std: Performing General Enumeration against: networkwalks.com...
2026-11-09T04:08:08.313979-0500 ERROR No answer for DNSSEC query for networkwalks.com
2026-11-09T04:08:08.928600-0500 INFO     SOA ns6135.hostgator.com 50.87.144.87
2026-11-09T04:08:09.719544-0500 INFO     NS ns6135.hostgator.com 50.87.144.87
2026-11-09T04:08:10.375997-0500 INFO     Bind Version for 50.87.144.87 "9.16.23-RH"
2026-11-09T04:08:10.377349-0500 INFO     NS ns6136.hostgator.com 192.232.216.131
2026-11-09T04:08:10.978560-0500 INFO     Bind Version for 192.232.216.131 "9.16.23-RH"
2026-11-09T04:08:12.134275-0500 INFO     MX mail.networkwalks.com 192.232.216.135
2026-11-09T04:08:12.773448-0500 INFO     A networkwalks.com 192.232.216.135
2026-11-09T04:08:14.052843-0500 INFO     TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-11-09T04:08:14.053435-0500 INFO     TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-11-09T04:08:14.897507-0500 INFO Enumerating SRV Records
2026-11-09T04:08:19.041080-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.15 443
2026-11-09T04:08:19.041485-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.14 443
2026-11-09T04:08:19.041574-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.14 443
2026-11-09T04:08:19.041687-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.9 443
2026-11-09T04:08:19.041733-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.11 443
2026-11-09T04:08:19.041808-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.11 443
2026-11-09T04:08:19.041871-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.15 443
2026-11-09T04:08:19.041914-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.9 443
2026-11-09T04:08:19.043296-0500 INFO 8 Records Found
2026-11-09T04:08:19.043607-0500 INFO Completed enumeration for domain: networkwalks.com
```

Zenmap

Scan Tools Profile Help

Target: [redacted] Profile: Ping scan Scan Cancel

Command: nmap -sn [redacted]

Hosts Services

OS Host

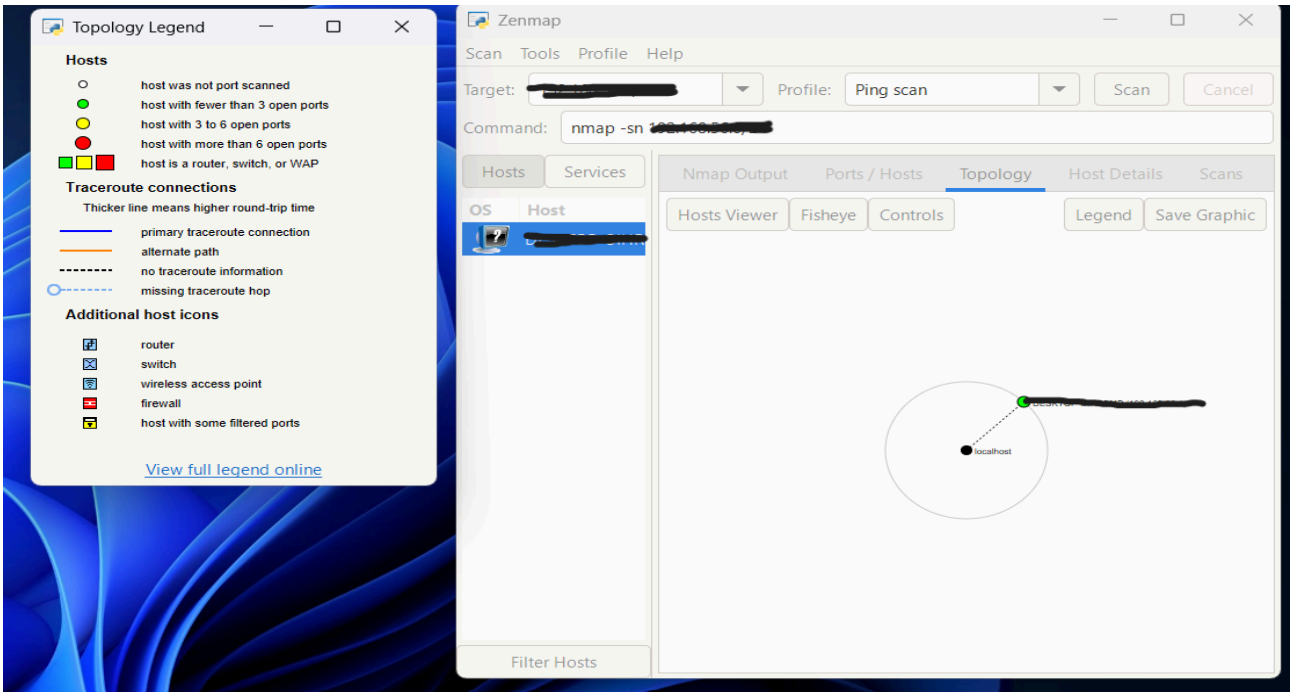
DESKTOP-[redacted]

Nmap Output Ports / Hosts Topology Host Details Scans

nmap -sn [redacted] Details

```
Starting Nmap 7.993 ( https://nmap.org ) at 2026-11-13
17:55 +0530
Nmap scan report for [redacted]
Host is up.
Nmap done: 256 IP addresses (1 host up) scanned in 12.75
seconds
```

Filter Hosts


Topology Legend
Hosts
host was not port scanned
host with fewer than 3 open ports
host with 3 to 6 open ports
host with more than 6 open ports
host is a router, switch, or WAP
Traceroute connections
Thicker line means higher round-trip time
primary traceroute connection
alternate path
no traceroute information
missing traceroute hop
Additional host icons
router
switch
wireless access point
firewall
host with some filtered ports
View full legend online
Zenmap
Scan Tools Profile Help
Target:
Profile: Ping scan
Scan
Cancel
Command: nmap -sn
Hosts
Services
OS Host
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
Hosts Viewer
Fisheye
Controls
Legend
Save Graphic
localhost
Filter Hosts

- End -